

MD TAKIT AHAMED

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RESEARCH PROFILE

Bachelor of Pharmacy student and research intern at Chang Gung University with experience in computational pharmacology, natural-product drug discovery, cancer cell-cycle research, comparative transcriptomics, and cellular senescence. My current work translates large scale screening results into experimentally actionable compound shortlists by integrating docking evidence, ADMET predictions, pharmaceutical developability, assay feasibility, and evidence confidence. I am developing a broader research focus on how genomic and metabolic stress produces persistent or reversible cell states, and how those states can be manipulated for cancer therapy and healthy aging. 4 peer reviewed publications

RESEARCH INTERESTS

DNA repair and genome stability, Cellular senescence and persistent stress states, Cancer treatment response, Natural product pharmacology, Healthy aging and geroscience, Comparative transcriptomics, Computational drug discovery

EDUCATION

Bachelor of Pharmacy (B.Pharm.), Universitas Sebelas Maret (UNS), Surakarta, Indonesia **Expected Dec. 2026**

- Undergraduate thesis: "Integrated Computational Prioritization of *Phyllanthus emblica* Phytochemicals Targeting EGFR and VEGFR-2 in Non-Small Cell Lung Cancer"
- Integrated phytochemical curation, target and pathway mapping, molecular docking, ADMET/drug-likeness assessment, and molecular dynamics to prioritize mechanistically plausible NSCLC leads.

RESEARCH EXPERIENCE

Research Intern, Yeast Biochemistry and Genetics Laboratory, Chang Gung University, Taiwan **Aug 2026 - Present**

APC/C-CDC20 small molecule prioritization and translational cancer research

- Prioritize small molecules identified from APC/C docking screens, integrating site level docking evidence, ligand-efficiency sensitivity analysis, ADMET predictions, assay exposure feasibility, chemical liabilities, and pharmaceutical developability.
- Develop and audit structured candidate registries and decision frameworks that connect computational screening results to experimentally actionable shortlists for MDA-MB-231 and RPE-1 validation.
- Evaluate compound specific uncertainty, formulation/vehicle constraints, predicted safety and metabolism, and evidence confidence to support final selection of candidates for downstream wet lab testing.

RESEARCH PROJECTS

Transcriptomic Aging Measurements During Cancer Cell Repopulation **June 2026-Ongoing**

- Implemented a reproducible Python/R workflow for quality control and transcriptomic aging score analysis of 45 RNA-seq libraries spanning control, therapy induced senescence, and repopulation conditions.
- Applied the Pasta transcriptomic aging model and evaluated within cell line contrasts, preprocessing sensitivity, and cell-cycle associations; extension to independent functional outcome datasets is ongoing.

Persistent DDR and Cellular Senescence Transcriptomic Atlas

Protocol development

Dec 2025-Present

- Developed a research protocol and separate evidence grading frameworks for DDR persistence (D0-D4) and senescence (S0-S4) to guide public dataset selection and phenotype annotation.
- Designed within study comparisons and a planned cross study meta analysis of shared and cell type specific transcriptional programs associated with persistent DDR and senescence.

Cupressus funebris Phytosteroids for Atopic Dermatitis Relief

Jan - Apr 2025

- Investigated phytosteroid rich extracts using Soxhlet extraction, purification/fractionation, topical formulation, and in vivo evaluation in a mouse model of induced atopic dermatitis.

PEER REVIEWED PUBLICATIONS

1. Ainurofiq, A., Maharani, E. P., Putranti, I. N. D., Wijaya, N. K., Putri, R. C. A., Maharani, V., & **Ahamed, M. T.** (2026). Recent advances in transdermal delivery: Correlating characteristics of nanoparticles with permeability in lipid-based

- drug delivery systems. Journal of Environmental Nanotechnology, 15(1), 340-348. <https://doi.org/10.13074/jent.2026.03.2612038>
2. Rizvi, H. H., Arshad, M. M., Sadia, S. H., Ali, I., **Ahamed, M. T.**, & Bibi, A. (2025). Pharmacological properties and clinical efficacy of *Ginkgo biloba* in cognitive decline and Alzheimer's disease. Journal of Medical & Health Sciences Review, 2(3), 4950-4962. <https://doi.org/10.62019/av8x4163>
 3. **Ahamed, M. T.**, Ramadhana, I. H. A., Purnama, M. F., & Ainurofiq, A. (2026). Antihypertensive herbal medicines as adjuncts to guideline-based care: A narrative review of clinical evidence, mechanisms, safety, and translational challenges. Advanced Drug Sciences, 1(1), e000004.
 4. Safa, Y. F., Ramadhani, S. N., Sabila, T. N., Sugiharti, S. N., Dharmawan, N. L., Faricha, E. A., Herawati, M., Nugrahayu, S. D., Barra, A. S. B., Maulidia, L. I., Reynanda, F. N., Dalena, R. M., **Ahamed, M. T.**, & Az Zahra, M. (2025). Implementasi edukasi swamedikasi dan pola hidup sehat dalam rangka pengabdian kepada masyarakat di RW 30 Dusun Randusari, Mojosongo, Jebres, Surakarta. Jurnal Pengabdian Masyarakat Bangsa, 3(5), 2651-2666. <https://doi.org/10.59837/jpmba.v3i5.2800>

MANUSCRIPTS IN PREPARATION

- **Ahamed, M. T.** Sari, Putri K. N, Amaliyah, A. and Mizranita, V. Targeting Tumor-Intrinsic Oncogenic Signaling and Angiogenesis in NSCLC: Integrated Computational Prioritization of *Phyllanthus emblica* Phytochemicals
- **Ahamed, M. T.** and Hossain M.S. Beyond γ H2AX: Defining Persistent DNA Damage Response in Cellular Senescence. (Examining how acute DDR, persistent DDR, and cellular senescence should be experimentally distinguished, emphasizing kinetics, retained damage foci, γ H2AX/53BP1 co-foci, DNA-SCARS, telomere dysfunction induced foci, and durable arrest)

TECHNICAL SKILLS

Computational & bioinformatics	Python; R; RDKit; network pharmacology; cheminformatics; comparative transcriptomics; KEGG and Gene Ontology analysis; protein-protein interaction analysis; ADMET prediction; molecular docking; molecular dynamics; BioRender.
Experimental & laboratory	Cell culture; DNA isolation; HPLC; UV-Vis spectrophotometry; Soxhlet extraction; phytochemical analysis; pharmaceutical formulation/technology; basic in vivo research exposure.
Research & analysis	Literature synthesis; evidence stratification; compound prioritization; pathway interpretation; multi-target mechanism mapping; hypothesis development; scientific writing; data standardization and decision framework design.

HONORS & SERVICE

Second Runner-Up, Bangladesh Regional Biology Olympiad - Bangladesh Biology Olympiad, 2020
Cadet - Bangladesh National Cadet Corps, 2017-2019

LANGUAGES

Bengali: Native | **English:** IELTS 6.5 | **Indonesian:** Working proficiency

ACADEMIC REFEREES

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